



Fig. 2: Koh Young's 8-way projection to overcome shadow problem

camera, AC servo control, XY work-bench, image processing system, and other actuating parts.

Illumination system

While older AOI machines used fluorescent, incandescent, ultraviolet, or infrared lights, today's machines use smart red, green, white, or blue coloured light emitting diodes (LEDs). LEDs are preferred for their ability to provide a stable form of lighting. With efficient control over the light, LEDs provide uniform illumination.

Along with the type of lighting, the orientation of the light source is also important. Machine vision camera systems assist the illumination system with effective lighting control and intensity adjustments to capture the entire PCB in audit.

CCD image system

CCD is a charge coupled device, a semiconductor element that converts images into digital signals. An AOI system uses a CCD camera for imaging. The system consists of a lens, an image sensor, and image processing tools.

The image sensor forms the heart of the camera. The photon energy captured by the camera lens and the individual pixel information is absorbed by the respective photocells in the sensor. The camera sensor converts incoming photon energy into electrical charges, which are further processed to generate images. The image processing system is the

key for extracting the information through images in any application.

In AOI, the processing of images takes place externally through the PC based system supported by various processing tools. The AOI machine's imaging resolution defines the amount of detail it can identify and capture, and hence it is a key factor that affects the inspection accuracy and inspection speed.

Machine vision camera

The image capture takes pictures for analysis. This is assisted by one or more high-definition cameras installed within the AOI to focus or move under the software command. Recently, these cameras have been featured with higher frame rate to support faster scanning and generate more data. An efficient processing software is hence required to manage huge amount of data.

Resolution turns out to be an important aspect as it defines the field of view (FOV) that specifically shows the section in an image. A wider FOV would typically allow the product to be examined with lesser images. However, it takes more time to capture each image as the camera with more pixels leads to a lower frame rate. On the other hand, cameras with a narrow FOV have bigger pixels with higher frame rate.

Processing software

For an AOI machine to inspect a product, the data of an acceptable

product must already be fed to the system. There are primarily two ways in which AOI machines are programmed. Image based programming uses scanned images of 'golden board' where references are created and accordingly automated. Algorithm based programming is the other method where the system generates its profile for the product by feeding product data to it.

Now, with all the elements working together, systematic outputs are obtained. Depending on the results, corrections are either made automatically or by human interference.

AOI challenges

While there have been great developments in the manufacturing of AOI machines over the years, many challenges still persist. Manufacturers are coming up with various solutions that are described below to tackle them.

Optical inspection. AOI systems were earlier based on 2D inspection technology where different grey levels for solder paste, solder joint, and component detection were inspected. Today, the 2D technology is being overtaken by 3D measurement solutions.

While many 2D AOI systems remain in use, the effort needed to keep this technology at a low level of escape and false-call rates is tremendously high. This is because the 2D systems work by comparing reference images. To help, some equipment manufacturers have added additional cameras and projectors to create quasi 3D or 2.5D technology, but it is still based on the same antiquated, inadequate comparison, or colour assessment.

AOI machines enabling both 2D and 3D imaging features along with other capabilities are being developed to provide extensive precision in the inspection process. Today, most AOI equipment has the capability to perform multiple tasks, reducing the time and expenses in the SMT process.